

Some Solutions for Chapter 2

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Feb. 21, 2006

- 2.11 (a) Our linear equations are, dropping primes and using the notation $\partial_x u = \partial u / \partial x$, etc.,

$$\partial_t u = -\partial_x \phi \quad (1)$$

$$\alpha^2 \partial_t w = -\partial_z \phi + b \quad (2)$$

$$\partial_x u + \partial_z w = 0 \quad (3)$$

$$\partial_t b + w N^2 = 0. \quad (4)$$

Rather than reduce things to a single equation for w and then assume sinusoidal solutions, let's do it the other way around just for a change. Assume that u, w, ϕ, b all vary as $\exp[i(kx + mz - \omega t)]$, and substitute in to get

$$-i\omega u = -ik\phi \quad (5)$$

$$-i\omega \alpha^2 w = -im\phi + b \quad (6)$$

$$iku + imw = 0 \quad (7)$$

$$-i\omega b + N^2 w = 0. \quad (8)$$

Eliminating all other variables in favor of w (or any other variable for that matter) leads to the dispersion relation

$$\omega^2 = \frac{N^2 k^2}{\alpha^2 k^2 + m^2}. \quad (9)$$

Note that the limiting value of the frequency as $k^2/m^2 \rightarrow \infty$ is N/α . We can see just by looking at the dispersion relation that it will approach this limit faster for larger α , though plotting makes it explicit.

- 2.14 Our equations are

$$\frac{d\mathbf{v}}{dt} = -\frac{\nabla p}{\rho_0} + \mathbf{k}b + \nu \nabla^2 \mathbf{v} \quad (10)$$

$$\frac{db}{dt} = Q + \kappa \nabla^2 b. \quad (11)$$

We will assume here that μ and κ are constants. Following the book, we write

$$\frac{d\Phi}{dt} = -w,$$

and using that together with the buoyancy equation gives

$$\frac{d(b\Phi)}{dt} = -wb + Q\Phi + \kappa\Phi\nabla^2 b. \quad (12)$$

Then we take $\mathbf{v} \cdot$ the momentum equation to get the kinetic energy equation,

$$\frac{d(|\mathbf{v}|^2/2)}{dt} = bw - \nabla \cdot (\phi \mathbf{v}) + \mathbf{v} \cdot \nu \nabla^2 \mathbf{v}. \quad (13)$$

Adding (12) and (13) gives the dissipative analog to (2.112) in Vallis:

$$\frac{\partial}{\partial t} \left(\frac{1}{2} \mathbf{v}^2 + b\Phi \right) + \nabla \cdot [\mathbf{v} \left(\frac{1}{2} \mathbf{v}^2 + b\Phi + \phi \right)] = Q\Phi + \kappa\Phi\nabla^2 b + \mathbf{v} \cdot \nu \nabla^2 \mathbf{v}. \quad (14)$$

Let's work on the last term, which is the velocity dotted with the viscous force. Let's rewrite things in index notation. The viscous force is the divergence of the stress tensor:

$$\mu \nabla^2 \mathbf{v} = \nabla \cdot \boldsymbol{\tau}_{ij} = \frac{\partial}{\partial x_j} (2\mu e_{ij}), \quad (15)$$

where we've used Einstein sum notation (sum over any index that appears twice), defined the dynamic viscosity $\mu = \rho_0 \nu$, and defined the stress tensor,

$$e_{ij} = \frac{1}{2} \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right).$$

Note that we have used incompressibility in (15) in that

$$\frac{\partial}{\partial x_j} (2e_{ij}) = \frac{\partial}{\partial x_j} \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right) = \frac{\partial^2 v_i}{\partial x_j \partial x_j} + \frac{\partial^2 v_j}{\partial x_j \partial x_i},$$

but

$$\frac{\partial^2 u_j}{\partial x_j \partial x_i} = \frac{\partial}{\partial x_i} \left(\frac{\partial u_j}{\partial x_j} \right) = 0,$$

since, translating back to vector notation, $\frac{\partial v_j}{\partial x_j} = \nabla \cdot \mathbf{v} = 0$. Anyway, getting back on our train of thought, the last term on the RHS of (13) can be written

$$2\mu v_i \frac{\partial}{\partial x_j} (e_{ij}) = 2\mu \left[\frac{\partial}{\partial x_j} (v_i e_{ij}) - e_{ij} \frac{\partial v_i}{\partial x_j} \right].$$

Now the so-called velocity gradient tensor, $\partial v_i / \partial x_j$, can be written as the sum of its symmetric and anti-symmetric parts,

$$\frac{\partial v_i}{\partial x_j} = e_{ij} + r_{ij},$$

where r_{ij} is the anti-symmetric part. But the doubly contracted product of a symmetric and anti-symmetric tensor is zero, so

$$e_{ij} \frac{\partial v_i}{\partial x_j} = e_{ij}(e_{ij} + r_{ij}) = e_{ij}e_{ij}.$$

Thus, dividing by ρ_0 , we have

$$\mathbf{v} \cdot \nu \nabla^2 \mathbf{v} = 2\nu v_i \frac{\partial}{\partial x_j} (e_{ij}) = 2\nu \left[\frac{\partial}{\partial x_j} (v_i e_{ij}) - e_{ij} e_{ij} \right].$$

Now we're getting somewhere. The first term in brackets on the RHS is the divergence of a flux. If we integrate over a closed volume, we can use Gauss' law to convert it to a surface integral, and show that that surface integral vanishes, because it has the velocity in it, and for a motionless boundary in a viscous fluid (requiring a no-slip boundary condition), all three components of the velocity vanish at the boundary.

So, knowing that, let's now integrate (14) over a closed volume. The flux divergence on the left-hand side vanishes too, since it is just velocity times scalars. Let's assume statistical steady state, so we can drop the tendency term too. Then the balance is (in mixed notation)

$$0 = \int (\Phi Q + \kappa \Phi \nabla^2 b) dV - 2\nu \int e_{ij} e_{ij} dV.$$

The first term, containing contributions from the heating as well as diffusion of buoyancy, is the "buoyancy source". The second is the frictional dissipation, which is negative definite since the doubly contracted product $e_{ij}e_{ij}$ is a sum of squares of real quantities. So, we have solved the first part of the problem.

The solution to the second part of the problem is essentially given in section 15.2 and 15.2.2 in the book. A slightly different treatment follows. We just have to show that the heating has to occur at a lower level than the cooling in order for a steady state dissipation to be maintained. Let's consider just the true heating term, ΦQ ; we'll address the buoyancy diffusion term after. In a sense the result is obvious by inspection; since $\Phi = -z$, we require that on average Q be anti-correlated with z in order to make $\int \Phi Q dV$ positive. For example, just to make things concrete, let's say we have a heating that varies linearly in z ,

$$Q = Q_0(0.5 \pm z/z_T), \quad (16)$$

where our domain boundaries are at $z = 0$ and $z = z_T$. Then we have

$$\int \Phi Q dV = Q_0 A \int \Phi Q dz = Q_0 A (z_T^2/4 \pm z_T^2/3),$$

where A is the horizontal area of the domain. The integral will be positive or negative depending whether we choose the plus or minus sign in (16), i.e.,

whether the heating increases or decreases with height. You can get the same result if you choose a sinusoidal heating, etc.

Now what about the buoyancy diffusion term? Is there some way to reduce that to something with a unique sign, in general? I don't believe so. Just think about it physically. If we have a situation with stable stratification — light fluid on top of heavy fluid — then diffusion of density, as it tries to iron out the density distribution and bring it closer to a state of uniform density, will actually increase the amount of potential energy in the fluid, by making the light fluid at the top heavier and vice versa. Now, if we imagine our buoyancy-driven circulation forced by heating below cooling, in that case it may be possible to show that the stratification on average is unstable, and thus that the buoyancy diffusion is stabilizing (i.e. a potential energy sink), but let's not try to prove that here.

- 2.16 (a) Assuming that the scaling is that for geostrophic flow (it is not necessary to assume exact geostrophic balance here, only that the pressure and Coriolis terms have the same order of magnitude) we have

$$fU \sim \phi/L,$$

so $\phi \sim fUL$.

- (b) Taking the divergence of the geostrophic wind gives zero. Now in the Boussinesq system, mass conservation is

$$\nabla \cdot \mathbf{v} = 0,$$

with $\mathbf{v}$ the 3D velocity. But if the horizontal velocity is geostrophic, then we have

$$\nabla_z \cdot \mathbf{u} = 0,$$

which implies that

$$\frac{\partial w}{\partial z} = 0,$$

and thus that if we have a solid boundary where $w = 0$, then we must have $w = 0$ everywhere. The naive scaling estimate $W/H \sim U/L$ is clearly an overestimate. For flows with $Ro \ll 1$, then, which are close to geostrophically balanced, we expect that it will still be an overestimate, even if the divergence is not exactly zero.

- (c) Thus, we expect hydrostatic balance to be more accurate in rapidly rotating flows than non-rotating flows, since hydrostatic balance holds to the extent that vertical acceleration (which scale as vertical velocity, for a given time scale, e.g. advective) is small compared to pressure gradients and gravity.

(d) Extra credit: The advective term scales as Ro compared to the Coriolis and pressure gradient terms. From this it is straightforward to show that $W \sim RoUH/L$, i.e. smaller than the naive scaling by a factor Ro .

2.17 Let's use the version in log-pressure coordinates (though we didn't discuss it in class), eq. (2.209). The same result can be obtained (with slightly more calculation) from pressure coordinates; even the Boussinesq approximation, though inaccurate, will be off by less than an order of magnitude. But anyway in log-p we have

$$f \frac{\partial u_g}{\partial z} = -\frac{R}{H} \frac{\partial T}{\partial y},$$

where now z is log-pressure height, but that is close enough to geometric height to ignore the difference. Let's use $H = 7km$, $R = 287 J kg^{-1} K^{-1}$, and take the pole-equator temperature difference to be $60^\circ K$ in winter. The pole-equator distance is $\pi a_e/2 \approx 10,000km$, so the mean temperature gradient is $60/10^7 = 6 \times 10^{-6} K m^{-1}$. A midlatitude value for f is $10^{-4} s^{-1}$. Plugging in the numbers gives

$$\frac{\partial u_g}{\partial z} \approx 2.5 m s^{-1} km^{-1},$$

Then, as a crude first approximation let's assume there is zero wind at the ground (due to frictional drag there) and that the shear is uniform with height, so that we can just write

$$u(z) = \frac{\partial u_g}{\partial z} z.$$

This gives about $12.5 m s^{-1}$. That is perhaps a bit slower than what is observed in winter in midlatitudes, but that is because the temperature gradients are not uniformly distributed in latitude but are concentrated in baroclinic zones. You can get a stronger wind by using the local temperature gradient in midlatitudes.

In summer, the pole-equator temperature difference is more like $30^\circ K$, so the same calculation gives $\sim 6 m s^{-1}$ at 5 km.

In the ocean, using the same pole-equator temperature differences but the Boussinesq thermal wind relation. We neglect salinity and assume a thermal expansion coefficient $\beta_T = 2 \times 10^{-4} K^{-1}$. Then a difference of 60K results in a pole-equator density difference of about 1.2%, or a buoyancy difference of that times g , or about $0.1 m s^{-2}$. Using (2.202), I get a shear on the order of $10 cm s^{-1} km^{-1}$. Notice that the problem doesn't ask for a typical current value at some depth... that's because unlike in the atmosphere, where the winds are weakest near the surface, in the ocean the surface has the largest velocities. The surface velocity is the integration constant we need to compute the velocity at some depth, so we can't compute it. Another complication is that the density gradient also drops off with depth relatively fast, also unlike the atmosphere.

2.18 In the atmosphere, we have to use the ideal gas expression

$$N^2 = \frac{g}{\theta} \frac{\partial \theta}{\partial z}.$$

If your only source of information is the temperature sounding I provided, this means you need to compute θ from T and p . For the troposphere, at the surface $\theta = T \approx 290K$, while at the tropopause $\theta \approx 220(1000/200)^\kappa \approx 350K$. Thus using a 60K drop over about 10km to estimate $\partial\theta/\partial z$, taking an intermediate value for θ itself of 320K, plugging in, and evaluating in terms of the period rather than frequency itself, I get $2\pi/N \approx 7.5$ minutes. For the stratosphere, it depends exactly which part of the stratosphere you choose, but using the lowest, isothermal layer I get about 5 minutes.

In the oceanic thermocline, the potential density varies by about 3 kg m^{-3} over about 200m. Using that to estimate the effective $\partial\rho/\partial z$, and using

$$N = \sqrt{-\frac{g}{\rho} \frac{\partial \rho}{\partial z}},$$

with a mean density of 1000 kg m^{-3} , I get about 8.5 minutes. Isn't it somewhat interesting that the ocean and atmosphere values are so similar? Why should that be? For the abyss, I get a drop of about 0.4 kg m^{-3} over about 4000m. Plugging that in gives $2\pi/N \approx 105$ minutes, or almost a couple hours. Our parcels will oscillate rather sluggishly down here.